FINANCIAL ANALYSIS PLANNING & FORECASTING FINAL PROJECT

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I HAVE NEITHER GIVEN NOR RECEIVED HELP (APART FROM THE INSTRUCTOR) TO COMPLETE THIS ASSIGNMENT

ABSTRACT

For the 2021-2022 regular season, the NBA released a list of the top 75 players of all time, voted on by an undeniably credible panel ([1]NBA 75th Anniversary Team). Using this list of the greatest players of all time, I tried to predict the number of rings a player would win in his career.

I used numerical factors such as players' career points per game average, playoff appearances, seasons played, Most Valuable Player awards won, how many times they were 1st team All-NBA([2]Basketball Reference All-Defense Historical Data)([3]Stat Muse)([4]Basketball Reference). I also included categorical factors such as the position each player played, or the time period they competed in for most of their career. Overall, this is a collection of factors I think make an impact on how many rings a player can win in their career!

In the first test of this study used ordinary least squares regression on a multivariable model to find significant factors in the number of rings a player wins. When I reran the test only including those significant variables, all variables remained significant with minor shifts.

In conclusion, I found career field goal percentage, wins, losses and competing in the Eastern conference may be significant in the number of rings a player will win in his career. Although due to assumption violations, I think a more accurate test can be made with time period specific data or with alternative regression and machine learning methods.

BACKGROUND INFORMATION

- In the National Basketball Association (NBA), no achievement is deemed greater than winning a championship ring. When debating who is the greatest basketball player, people often immediately turn to how many rings they have won in their NBA career as an argument. Only 1 team earns it per year, and it has eluded some of the greatest basketball players of all time. However, it is interesting that some of the players that made the top 75 list have never won a championship ring. As a reminder, this is an undeniably credible panel which includes former and current NBA players, NBA coaches, WNBA coaches and players, and sportswriters who have covered NBA basketball for decades.
- A very valuable debate and conversation topic would be the use of this tool and forecasting just how successful a player will be by the number of championships they win. This is not predicting the potential of greatness in a player, because the list proves you do not need a championship ring to be great. However, this is a new tool that can be used as a benchmark or performance measure for future generations of NBA players to come! General Managers can reference this when drafting a rookie or signing a free agent onto their team. Players can use this when they're seeking new teammates to help them win a championship. This is an invaluable tool for people at all levels in the NBA.

DESCRIPTION OF THE FACTORS

- Player
  - Each NBA Player that was selected to the top 75 list.
- Total Rings (Y)
  - The number of rings earned by an NBA player in his career
- Career PPG
  - The average points scored per game over the career of a player
- Career APG
  - The average assists made per game over the career of a player
- Career RPG
  - The average rebounds per game over the career of a player
- Career SPG
  - The average steals per game over the career of a player
- Career BPG
  - The average blocked shots per game over the career of a player
- Career FG%
  - The overall shooting accuracy of a player over their career
- Career 3PFG%
  - The shooting accuracy from the 3-point line (22 feet+) of a player over their career
- Career Points High
  - The most points scored in a game by a player in their career
- Career Assists High
  - The most assists made in a game by a player in their career
- Career Rebounds High
  - The most rebounds grabbed in a game by a player in their career
- Playoff Appearances
  - How many times a player made the playoffs (post season play) in his career
- All NBA 1st Team Selections
  - How many times a player was selected to the All-NBA 1st team which deems them as one of the 5 best players in the NBA that year.
- Career Wins
  - Total career wins by a player.
- Career Losses
  - Total career losses by a player.

DESCRIPTION OF THE FACTORS (CONT.)

- All NBA Defensive Team Selections
  - How many times a player was selected to the All-NBA defensive 1st team which deems them as one of the 5 best defenders in the NBA that year.
- Seasons Played
  - Total seasons a player competed in their career.
- All Star Selections
  - How many times a player is selected to the NBA All-Star game which deems them as one of the best players on their team.
- Total MVPs
  - How many times a player won the Most Valuable Player award in their career. This award is given to the best player in the NBA for that respective season.
- Conference
  - Players competed in either the West or East conference for most of their career. Only one team from each conference gets a chance to win a championship per year.
- Position
  - Players filled roles of either guard, forward or center.
- Time Period
  - Early Years: NBA Players who competed between approximately 1946-1966. The NBA fluctuated in small size during these years with an average size of 9 teams.
  - Expansion: NBA Players who competed between approximately 1966-1988 when the size of the NBA grew from 10 teams to 23 teams.
  - Modern: NBA Players who competed between approximately 1989 – present. The size of the NBA is currently at 30 teams and still looking at more expansion.

RESIDUAL ANALYSIS

- Majority if not all the residual plots exemplify randomness as there is no clear pattern to the predicted values
- The randomness in the residual plots helps support the Gauss Markov assumption of randomness and independence of error terms
- All residual plots are viewable in the appendix

- I found these residual plots interesting:
  - Career SPG, Career BPG, Eastern Conference competition, Forward Position, Center Position, Modern Time Period, Expansion Time Period
  - These plots are all very similar with predicted values and actual values in almost the same spot
  - This is more than just insignificance in the model, as other insignificant models did not resemble a similar plot – and Eastern Conference competition is significant in the model
  - This may be due to the presence of bias in some variables but also the input values of these variables all range between 0 and 2 which may be the cause.

TEST MODEL 1 SUMMARY

```
Residuals:
    Min      1Q  Median      3Q     Max 
-2.4764 -0.8647  0.0053  0.6926  3.6058 

Coefficients:
                                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)                            9.813387   3.770308   2.603 0.012020 *  
`Career PPG (X1)`                      0.011281   0.132114   0.085 0.932278    
`Career APG (X2)`                     -0.381842   0.266330  -1.434 0.157639    
`Career RPG (X3)`                     -0.199085   0.190345  -1.046 0.300438    
`Career SPG (X4)`                     -0.249253   0.546120  -0.456 0.649997    
`Career BPG (X5)`                     -0.364271   0.450284  -0.809 0.422211    
`Career FG% (X6)`                     -0.149471   0.066505  -2.248 0.028875 *  
`Career 3PFG% (X7)`                   -0.002003   0.020221  -0.099 0.921458    
`Career Points High(X8)`              -0.002843   0.038534  -0.074 0.941478    
`Career Assists High(X9)`              0.103462   0.100705   1.027 0.308998    
`Career Rebounds High(X10)`            0.054493   0.084156   0.648 0.520140    
`Playoff Appearances (X11)`           -0.219108   0.130887  -1.674 0.100131    
`All NBA 1st Team selections (X12)`   -0.120344   0.101869  -1.181 0.242833    
`Career Wins(X13)`                     0.012783   0.003627   3.525 0.000893 ***
`Career Losses(X14)`                  -0.021204   0.003980  -5.327 2.17e-06 ***
`All NBA Defensive Team Selections(X15)` 0.141921 0.087268  1.626 0.109941    
`Seasons Played (X16)`                 0.159064   0.235362   0.676 0.502144    
`All Star Selections (X17)`            0.086828   0.113831   0.763 0.449041    
`Total MVPs (X18)`                     0.330995   0.245076   1.351 0.182678    
`Conference-East(Dummy)(X19)`          0.986331   0.474630   2.078 0.042653 *  
`Position-Forward(Dummy)(X20)`        -0.004430   0.809964  -0.005 0.995657    
`Position-Center(Dummy)(X21)`          1.157808   1.273459   0.909 0.367448    
`Time Period-Expansion(Dummy)(X22)`   -0.425439   0.788928  -0.539 0.592008    
`Time Period-Modern(Dummy)(X23)`      -0.606162   0.881373  -0.688 0.494668    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.473 on 52 degrees of freedom
Multiple R-squared:  0.7265,	Adjusted R-squared:  0.6055 
F-statistic: 6.006 on 23 and 52 DF,  p-value: 4.394e-08
```

Significant Variables at the 0.05 significance level

- Career FG% (X6) P-value (0.02) < alpha (0.05)
- Career Wins(X13) P-value (0.00) < alpha (0.05)
- Career Losses(X14) P-value (0.00) < alpha (0.05)
- Conference-East (Dummy) (X19) P-value (0.04) < alpha (0.05)

Adjusted R^2 = 0.61; 61% of the variance in number of rings won (Y) is due to the multiple variables in the model (X_i)

MODEL 1 VS. GAUSS-MARKOV ASSUMPTIONS: DW TEST

N = 75 alpha = 0.05 k (independent variables) = 20
H0: a = 0 H1: a > 0
"a" represents the amount of correlation between error terms in the model E_t = a_E{t-1} + W_t
where:
E_t = ith error term
W_t =random variable
Ideally, in the Durbin Watson test we want correlation to equal 0 because this means our error terms are random

dL = 1.027 dU = 2.315 D-stat = 1.82
D-stat (1.82) is > dL (1.027) and < dU (2.315) therefore the test is inconclusive. There is not enough evidence to reject H0 or not reject H0

	Assumption Violation?	Outcome
Linearity	**No**	My general assumption follows the multivariable regression model.
Significant Variables (T Test)	**N/A**	At a 0.05 significance level, the significant variables are Career FG% (X6), Career Wins(X13), Career Losses(X14) and Eastern Conference Competition (X19).
Randomness and Independence of Observations (Durbin Watson Test & Residual Plots) H0: a = 0 H1: a > 0	**Inconclusive**	Using 75 observations (n) and 20 independent variables(k) as measurement, the dstatistic (1.82) is > dL critical value (1.027) and is < dU critical value (2.315). Therefore **the test is inconclusive as there is not enough evidence to reject H0 or not reject H0.** Randomness may or may not be present

```
lag Autocorrelation D-W Statistic p-value
  1      0.08309896      1.829056   0.438
Alternative hypothesis: rho != 0
```

MODEL 1 VS GAUSS-MARKOV ASSUMPTIONS: BP TEST

K(independent variables) = 23 alpha = 0.05
H0: a_1=a_2=...=a_23=0
H1: At least one value among (a_1,a_23) is not 0

"a_i" represents the parameter for each variable in the model with the original residuals as the dependent variables and original factors as independent variables

For the BP test, we want the variance in the error terms to all equal 0, because this means there is no bias and all error is distributed equally.

$X^2_{23,0.05}$= 35.172 BP-stat = 31.211
BP-stat (31.211) is < $X^2_{23,0.05}$(35.172), therefore we DO NOT reject the null hypothesis. There is sufficient evidence that heteroscedasticity is not present in linear form

No Perfect Multicollinearity	No	Each estimator of the parameter is able to be calculated without error.
Zero Conditional Mean	No	Sum of residuals = 0 (3.080869e-15).
Homoscedasticity of Error Terms (BP Test) H0: There's no conditional heteroscedasticity H1: There is conditional heteroscedasticity	No	With 23 independent variables (k) and significance level 0.05, BPStat (31.411) is < chi square critical value (35.172). Therefore, we **do not reject** H0 at the 0.05 significance level. There is not sufficient evidence of Heteroscedasticity in linear form.

```
        studentized Breusch-Pagan test

data:  model
BP = 31.411, df = 23, p-value = 0.113
```

MODEL 1 VS. GAUSS-MARKOV ASSUMPTIONS: WHITE TEST

alpha = 0.05
H0: a_i =0
H1: At least one value among a_i is not 0

"a_i" represents the parameter for each variable in the regression model with the original residuals as the dependent variables and quadratic original factors as independent variables

For the White test, we want the variance in the error terms to all equal 0, because this means there is no bias and all error is distributed equally.

Homoscedasticity of Error Terms (White Test) H0: a1=a2=...=aN=0 H1: At least one value among ai is not 0	No	With 22 independent varibales (k) and significance level of 0.05, Wstat (16.28) is < chi square critical value (33.924) so **we do not reject** H0 at the 0.05 significance level. There is not sufficient evidence of heteroscedasticity in quadratic form.
Normality of Error Terms	YES	The Histogram does not represent a normal distribution.

```
White's test results

Null hypothesis: Homoskedasticity of the residuals
Alternative hypothesis: Heteroskedasticity of the residuals
Test Statistic: 16.28
P-value: 0.000292
```

$X^2_{22,0.05}$ = 33.924 W-stat = 16.28
W-stat is < $X^2_{22,0.05}$ therefore we DO NOT reject the null hypothesis. There is sufficient evidence that heteroscedasticity is not present in quadratic form

TEST MODEL 2 SUMMARY

```
Residuals:
    Min      1Q  Median      3Q     Max 
-3.0971 -0.8767 -0.0928  0.9063  4.7825 

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)    
(Intercept)                 6.784607   2.398936   2.828  0.00608 ** 
`Career FG% (X6)`          -0.107701   0.049507  -2.175  0.03292 *  
`Career Wins(X13)`          0.011014   0.001373   8.022 1.52e-11 ***
`Career Losses(X14)`       -0.017772   0.002052  -8.662 9.89e-13 ***
`Conference-East(Dummy)(X19)` 1.423813   0.383110   3.716  0.00040 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.59 on 71 degrees of freedom
Multiple R-squared:  0.5647,	Adjusted R-squared:  0.5401 
F-statistic: 23.02 on 4 and 71 DF,  p-value: 3.172e-12
```

Significant Variables at the 0.05 significance level

- Career FG% (X6) P-value (0.03) < alpha (0.05)
- Career Wins(X13) P-value (0.00) < alpha (0.05)
- Career Losses(X14) P-value (0.00) < alpha (0.05)
- Conference-East (Dummy) (X19) P-value (0.00) < alpha (0.05)

Adjusted R^2 = 0.54; 54% of the variance in number of rings won (Y) is due to the multiple variables in the model (X_i)

Less of the variance in the data is due to the variables in the model

Conference-East (Dummy) became largely more significant, and Career Wins became slightly more significant

Career FG% and Career Losses both lost significance

MODEL 2 VS. GAUSS-MARKOV ASSUMPTIONS: DW TEST

N = 75 alpha = 0.05 k (independent variables) = 4
H0: a = 0 H1: a > 0

"a" represents the amount of correlation between error terms in the model E_t = a_E{t-1} + W_t where:
E_t = ith error term
W_t = random variable

Ideally, in the Durbin Watson test we want correlation to equal 0 because this means our error terms are random

dL = 1.51 dU = 1.74 D-stat = 2.02
D-stat (2.02) is > dU (1.027) therefore we DO NOT reject H0 at the 0.05 significance level. There's sufficient evidence that randomness may not be present

	Assumption Violation?	Outcome
Linearity	No	My general assumption follows the multivariable regression model
Randomness and Independence of Observations (Durbin Watson Test & Residual Plots) H0: a = 0 H1: a > 0	No	Using 75 observations (n) and 4 independent variables(k) as measurement, the dstatistic (2.02) is > dU critical value (1.74).Therefore we do not reject **H0, there's sufficient evidence that** randomness may not be present
No Perfect Multicollinearity	No	Each estimator of the parameter is able to be calculated without error.
Zero Conditional Mean	No	Sum of residuals = 0 (2.331468e-15).

```
lag Autocorrelation D-W Statistic p-value
  1     -0.02643415      2.028797   0.894
Alternative hypothesis: rho != 0
```

MODEL 2 VS. GAUSS-MARKOV ASSUMPTIONS: BP TEST

K(independent variables) = 4 alpha = 0.05
H0: a_1=a_2=a_3=a_4=0
H1: At least one value among (a_1,a_4) is not 0

"a_i" represents the parameters for each variable in the model with the original residuals as the dependent variables and original factors as independent variables

Homoscedasticity of Error Terms (BP Test) H0: There's no conditional heteroscedasticity H1: There is conditional heteroscedasticity	YES	With 4 independent variables (k) and significance level 0.05, BPStat (18.685) is > chi square critical value (9.488). Therefore, we reject H0 at the 0.05 significance level. There is sufficient evidence of Heteroscedasticity in linear form.

```
        studentized Breusch-Pagan test

data:  model2
BP = 18.685, df = 4, p-value = 0.0009064
```

For the BP test, we want the variance in the error terms to all equal 0, because this means there is no bias and all error is distributed equally.

$X^2_{4,0.05}$= 9.488 BP-stat = 18.685
BP-stat (18.685) is > $X^2_{4,0.05}$(9.488), therefore we reject the null hypothesis. There is sufficient evidence of heteroscedasticity in linear form

MODEL 2 VS. GAUSS-MARKOV ASSUMPTIONS: WHITE TEST

alpha = 0.05
H0: a_i=0
H1: At least one value among a_i is not 0

"a_i" represents the parameters for each variable in the model with the original residuals as the dependent variables and quadratic original factors as independent variables

For the White test, we want the variance in the error terms to all equal 0, because this means there is no bias and all error is distributed equally.

Homoscedasticity of Error Terms (White Test) H0: a1=a2=...=aN=0 H1: At least one value among ai is not 0	YES	With 3 independent varibales (k) and significance level of 0.05, Wstat (26.32) is > chi square critical value (7.815) so **we reject** H0 at the 0.05 significance level. There is sufficient evidence of heteroscedasticity in quadratic form.
Normality of Error Terms	YES	The Histogram does not represent a normal distribution.

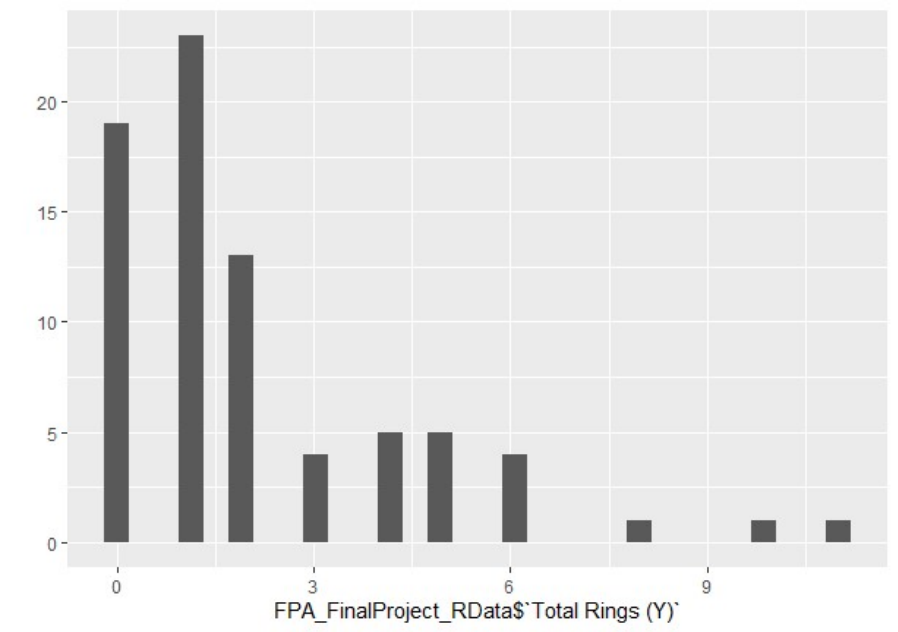
```
white's test results

Null hypothesis: Homoskedasticity of the residuals
Alternative hypothesis: Heteroskedasticity of the residuals
Test Statistic: 26.32
P-value: 2e-06
```

$X^2_{3,0.05}$ = 7.815 W-stat = 26.31
W-stat is > $X^2_{3,0.05}$ therefore we REJECT the null hypothesis. There is sufficient evidence that heteroscedasticity is present in quadratic form

GAUSS MARKOV ASSUMPTION VIOLATION RECOMMENDATIONS

- In both Model 1 and 2 the normality of error terms assumption is violated. That is proved by the histogram to the right not showing normal distribution
  - **For a future study, I would recommend using different types of regressions or machine learnings that do not require normal distributions.**
- Additionally, in model 2 the homoscedasticity of error terms assumption is violated in both the BP and White tests.
  - **This dataset considers data that was gathered over 75 years. Players competed at different levels and with different number of teams which can skew the data and introduce bias. I think to fix this, maybe data from one consistent time period will reduce bias.**



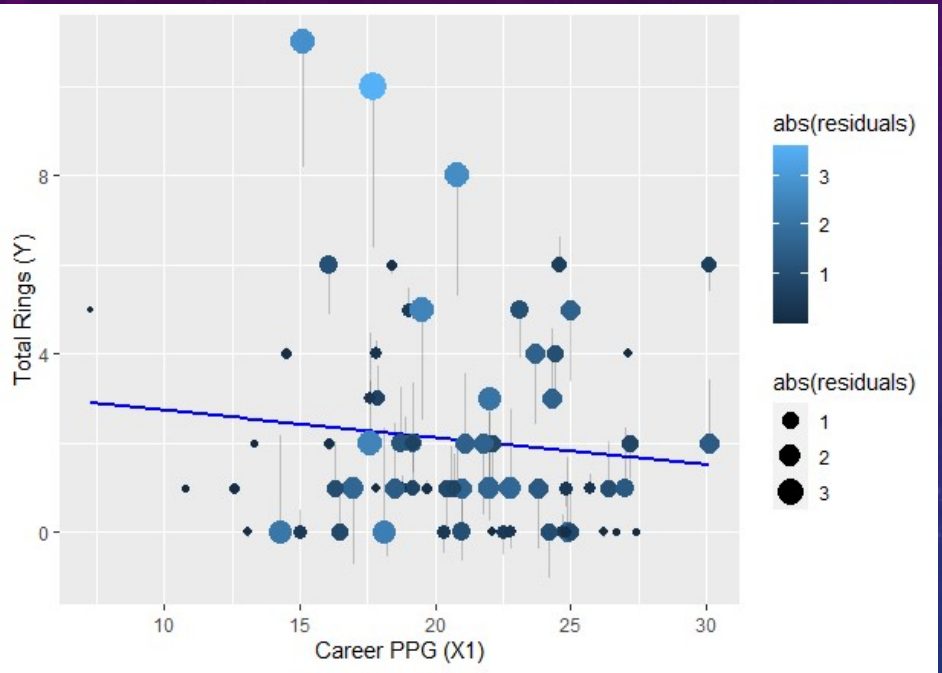
APPENDIX: WORKS CITED

- [1] NBA 75th Anniversary Official Top 75 List : https://www.nba.com/news/nba-75th-anniversary-team-announced
- [2]NBA All-Defensive Team Historical Data : https://www.basketball-reference.com/awards/all_defense_by_player.html
- [3]Statmuse NBA Career Stats : https://www.statmuse.com/nba
- [4]Basketball Reference NBA Career Stats : https://www.basketball-reference.com/awards/nba_75th_anniversary.html

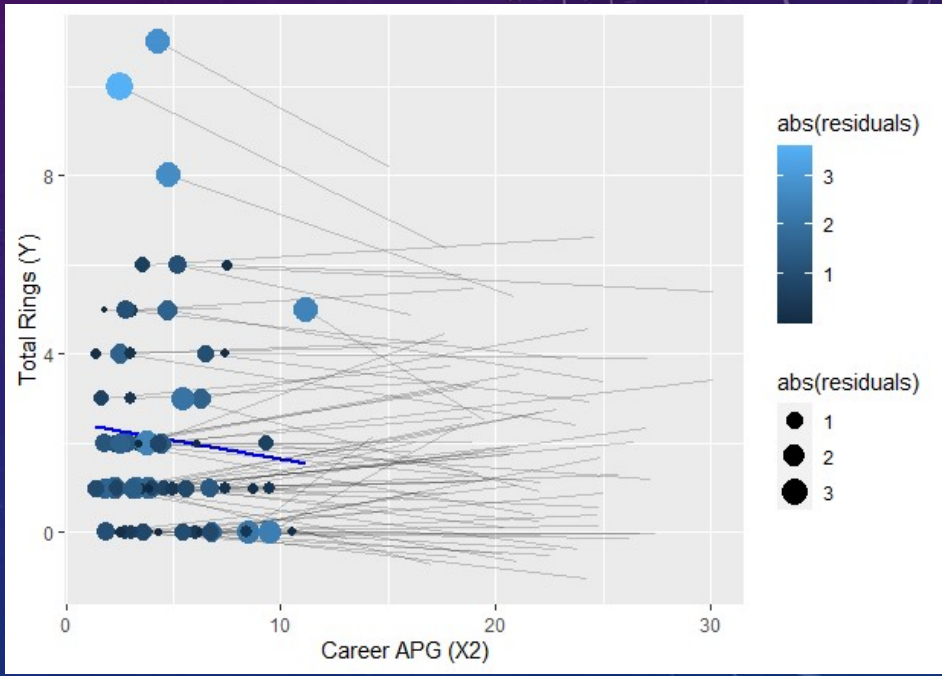
APPENDIX: RESIDUAL PLOTS

- These are residual plots for each variable
- The actual value of each observation is shown by a circle on the plot
- There is a line attached to each circle, leading to where the predicted value is
- Each point on the plot varies in brightness and size
- The **Brighter** a circle is, symbolizes the how great the error of that predicted value is
- The **Larger** a circle is, further symbolizes how great the error of that predicted value is

APPENDIX: RESIDUAL PLOTS CONT.

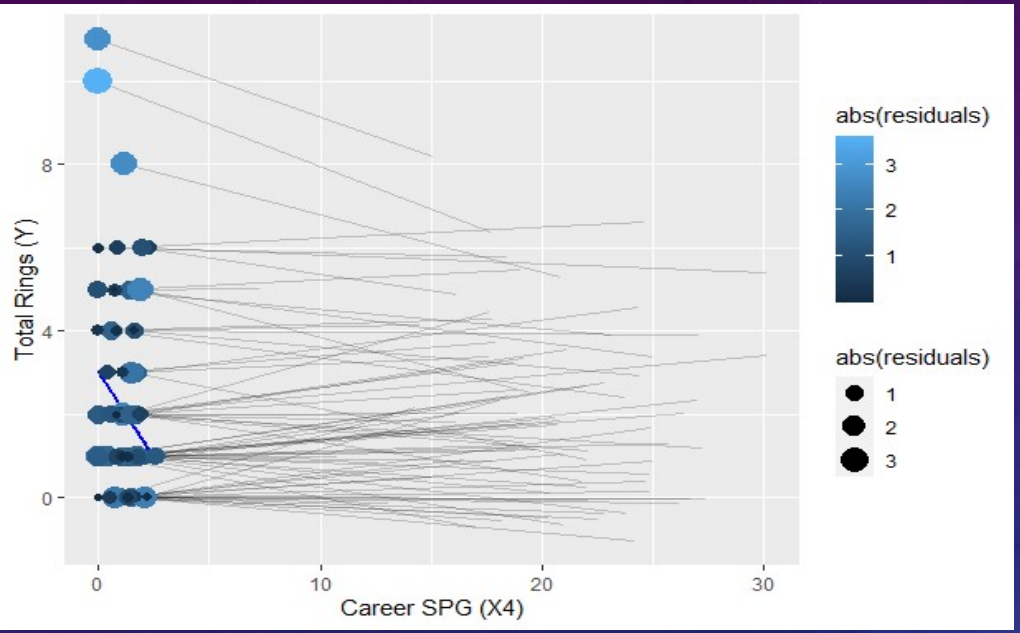


Career PPG

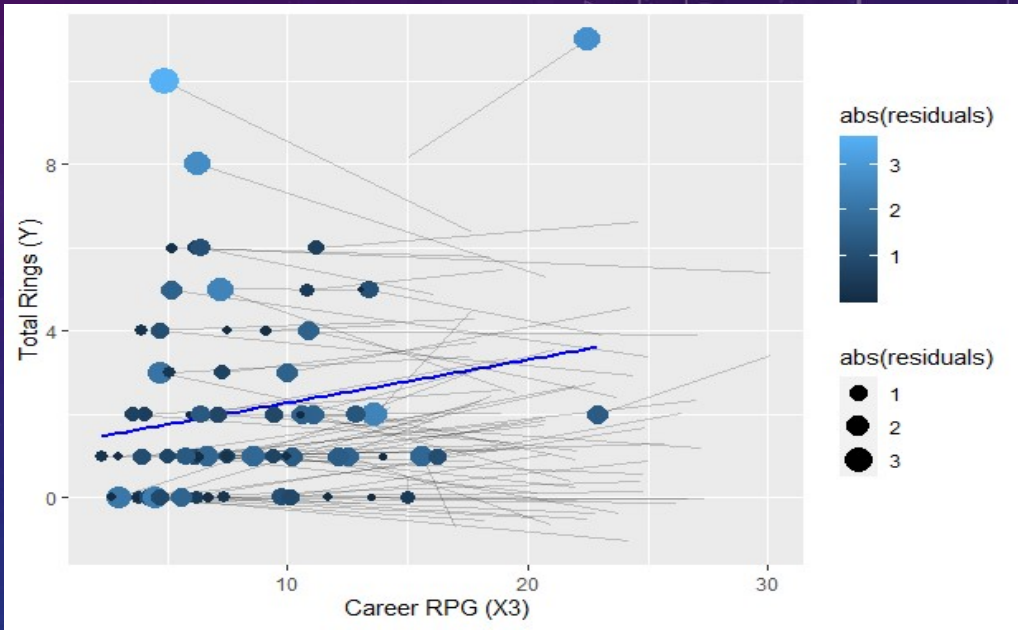


Career APG

APPENDIX: RESIDUAL PLOTS CONT.

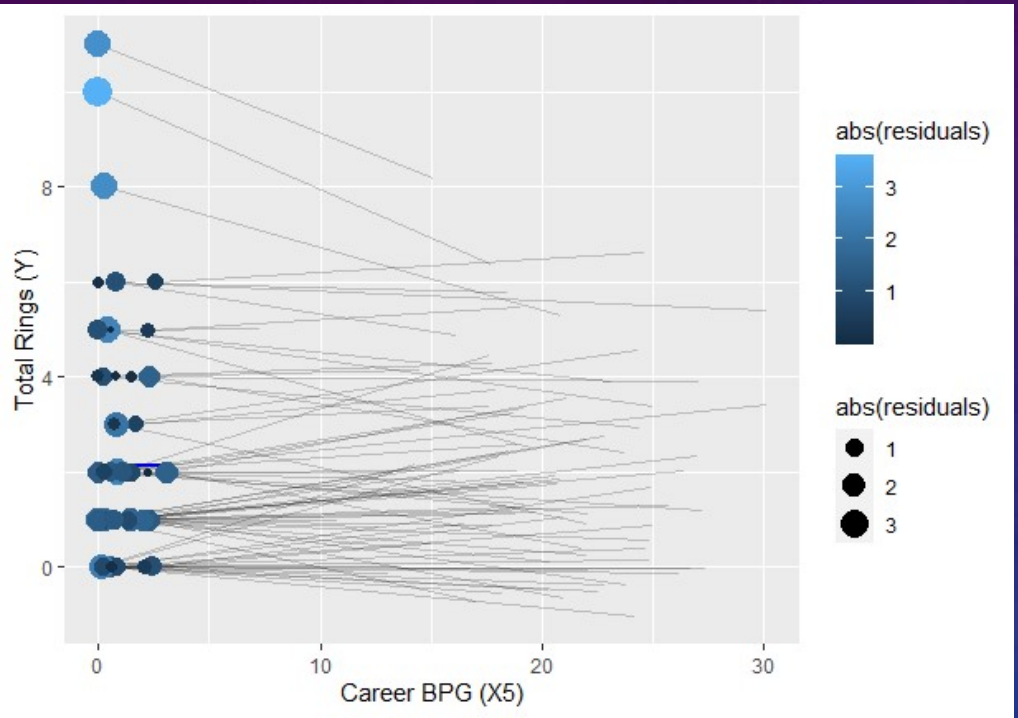


Career SPG

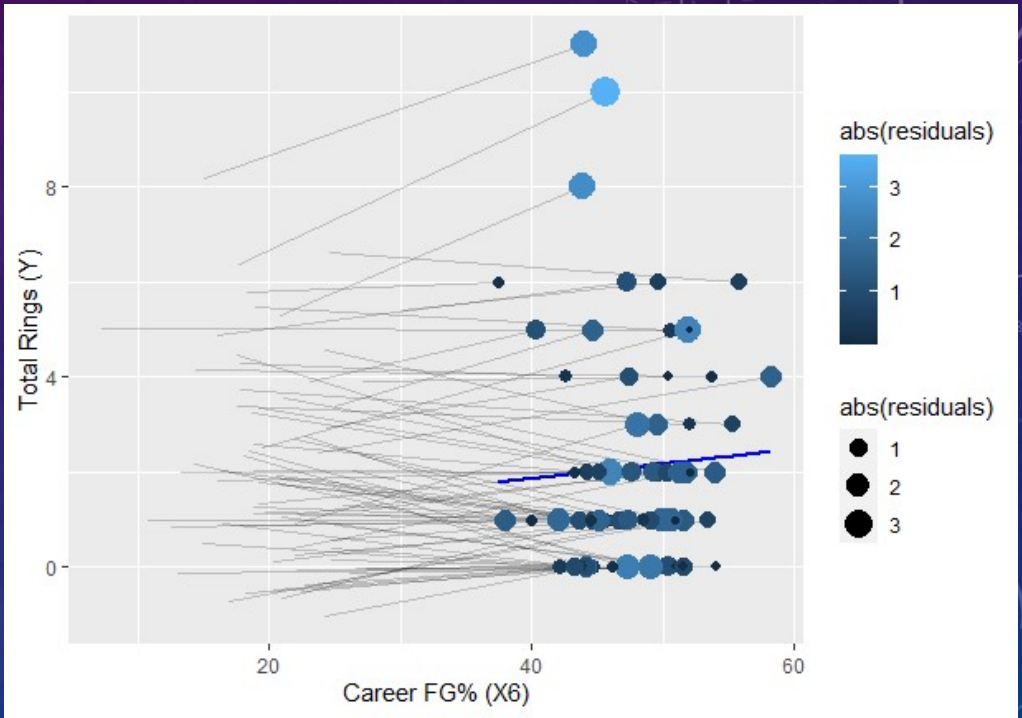


Career RPG

APPENDIX: RESIDUAL PLOT CONT.

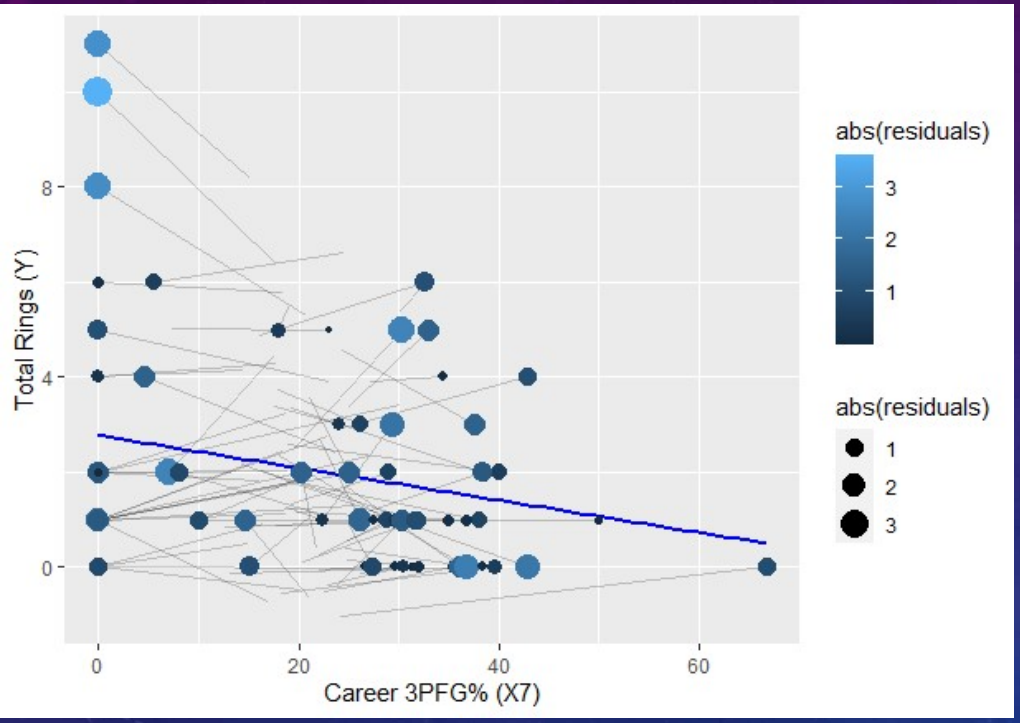


Career BPG

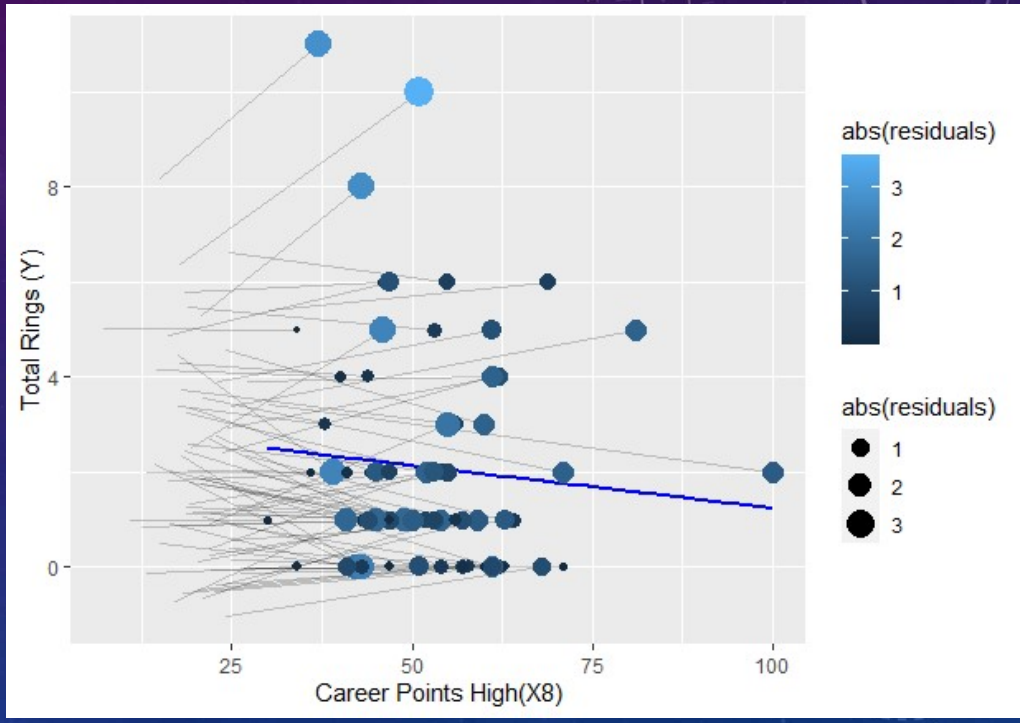


Career FG%

APPENDIX: RESIDUAL PLOT CONT.

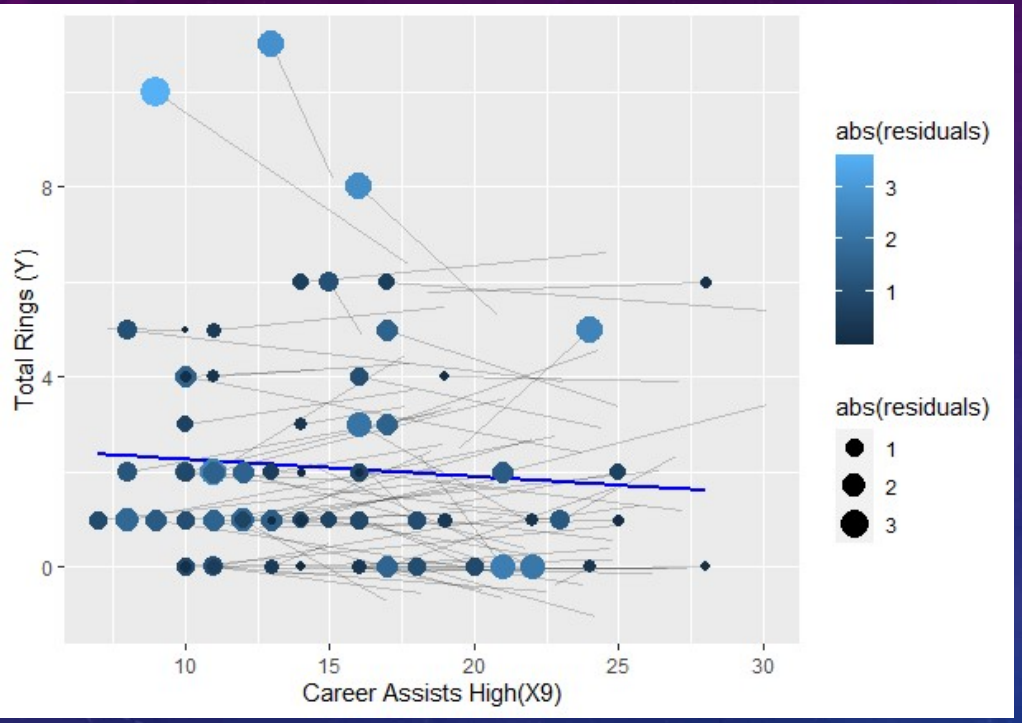


Career 3PFG%

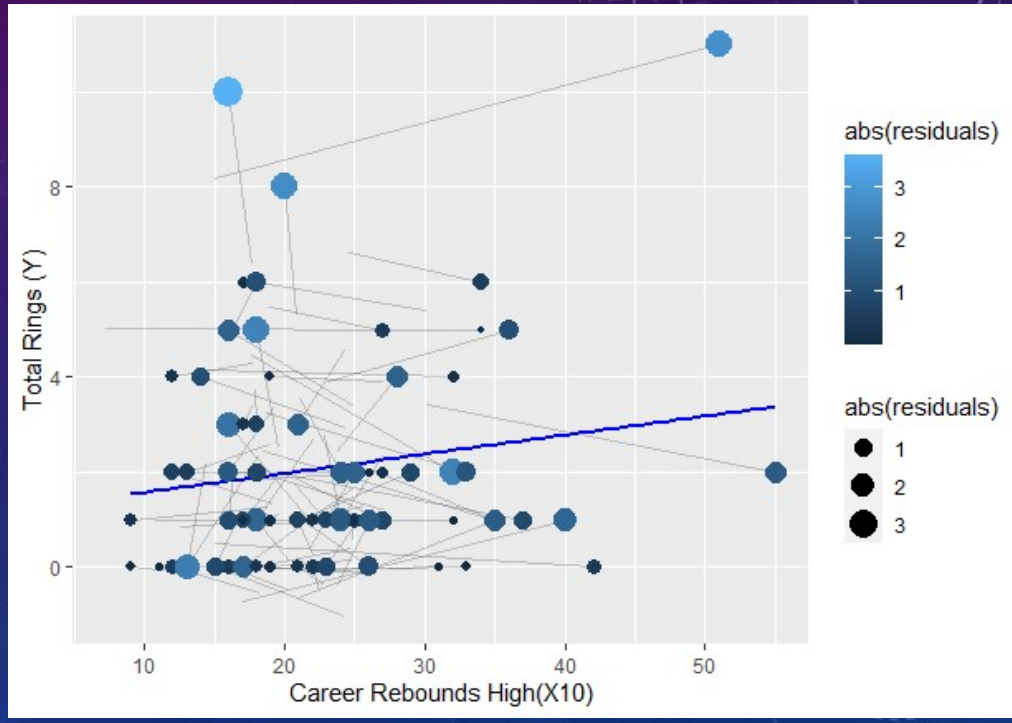


Career Points High

APPENDIX: RESIDUAL PLOT CONT.

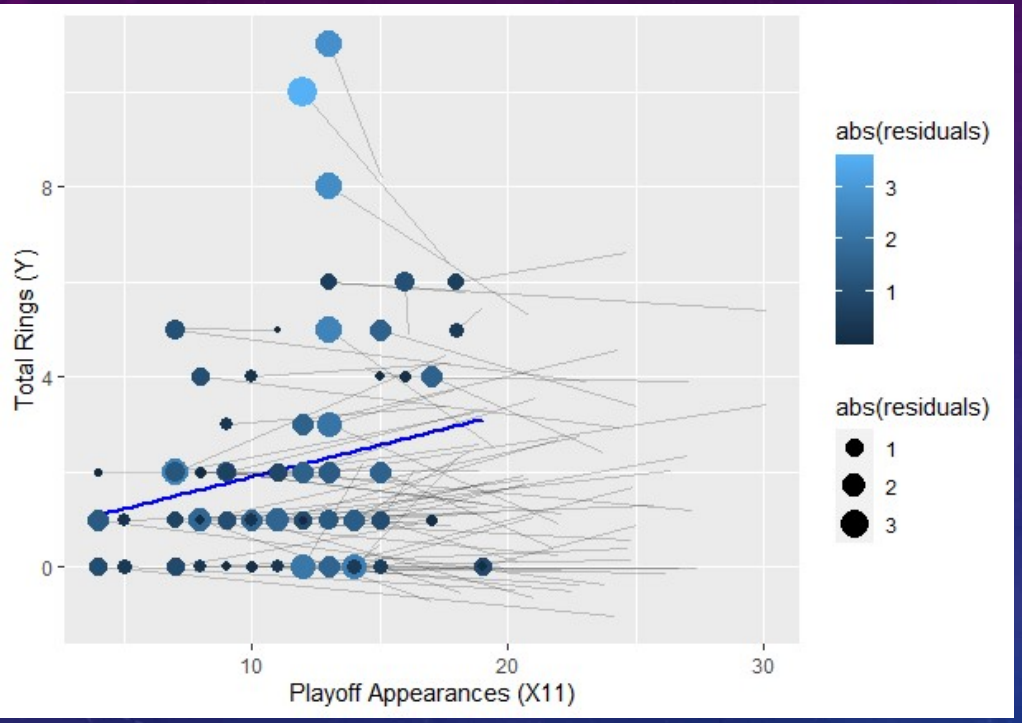


Career Assists High

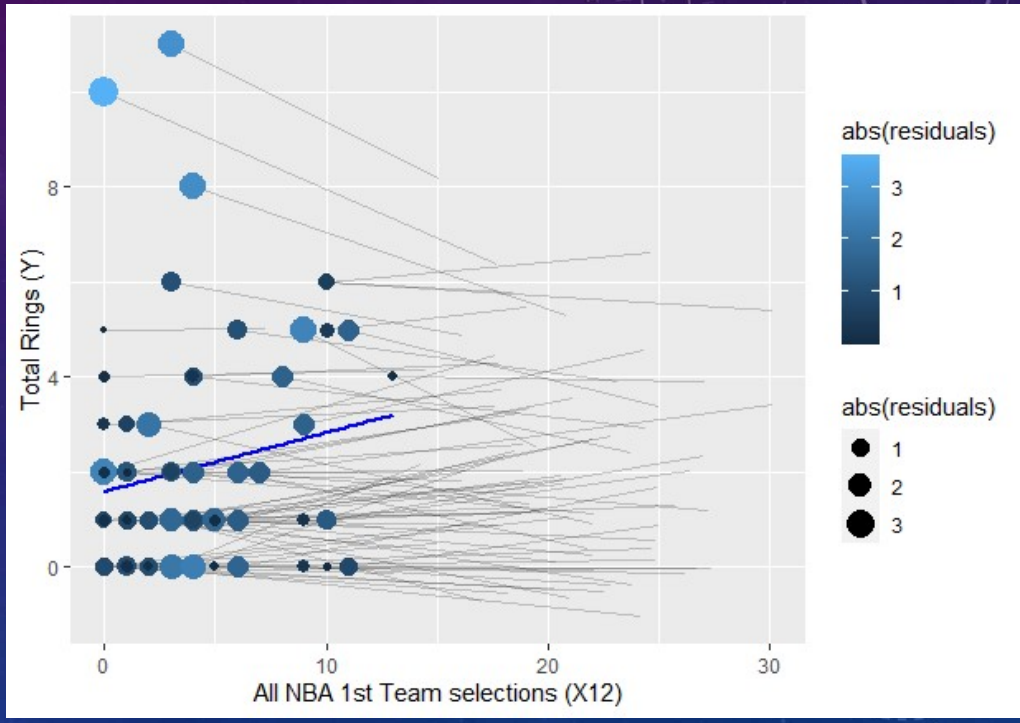


Career Rebounds High

APPENDIX: RESIDUAL PLOT CONT.

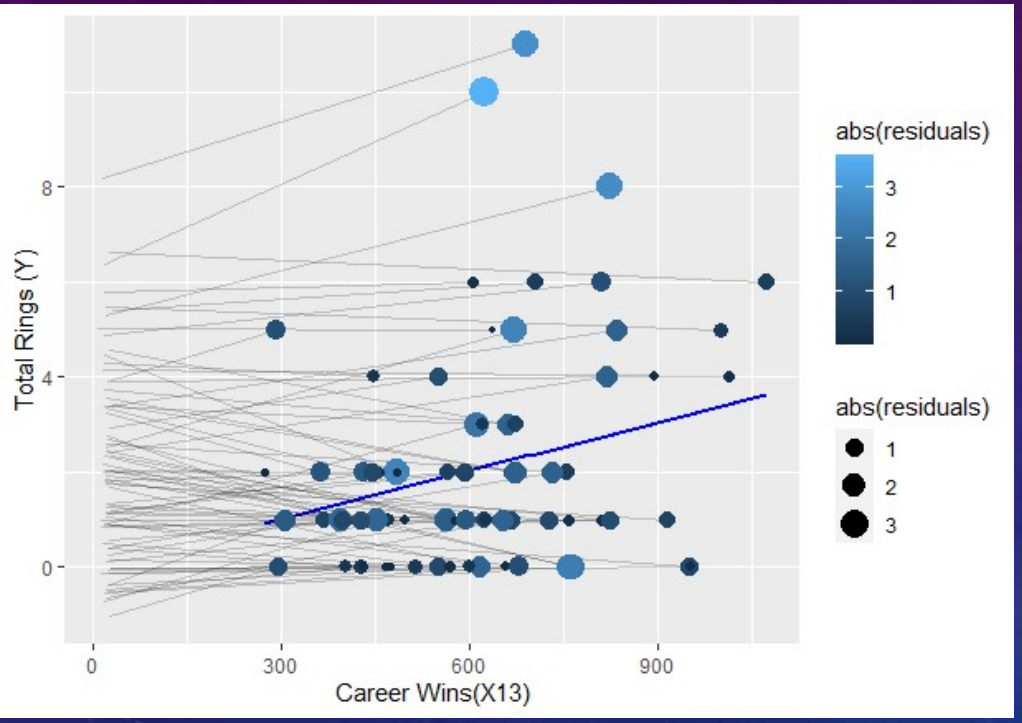


Playoff Appearances

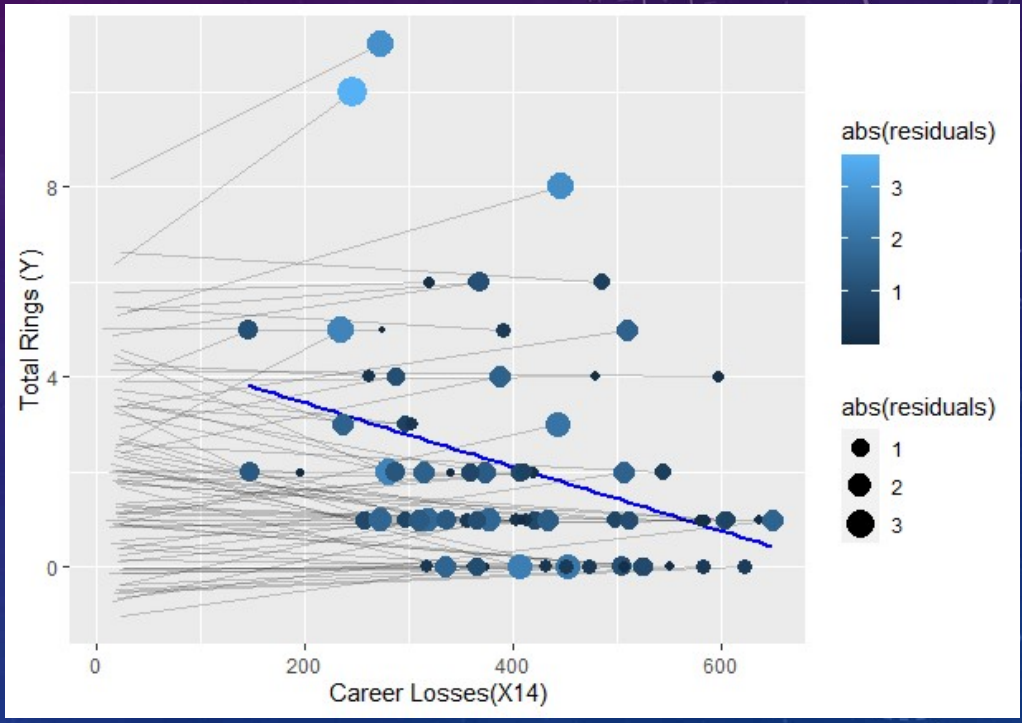


All NBA 1st Team Selections

APPENDIX: RESIDUAL PLOT CONT.

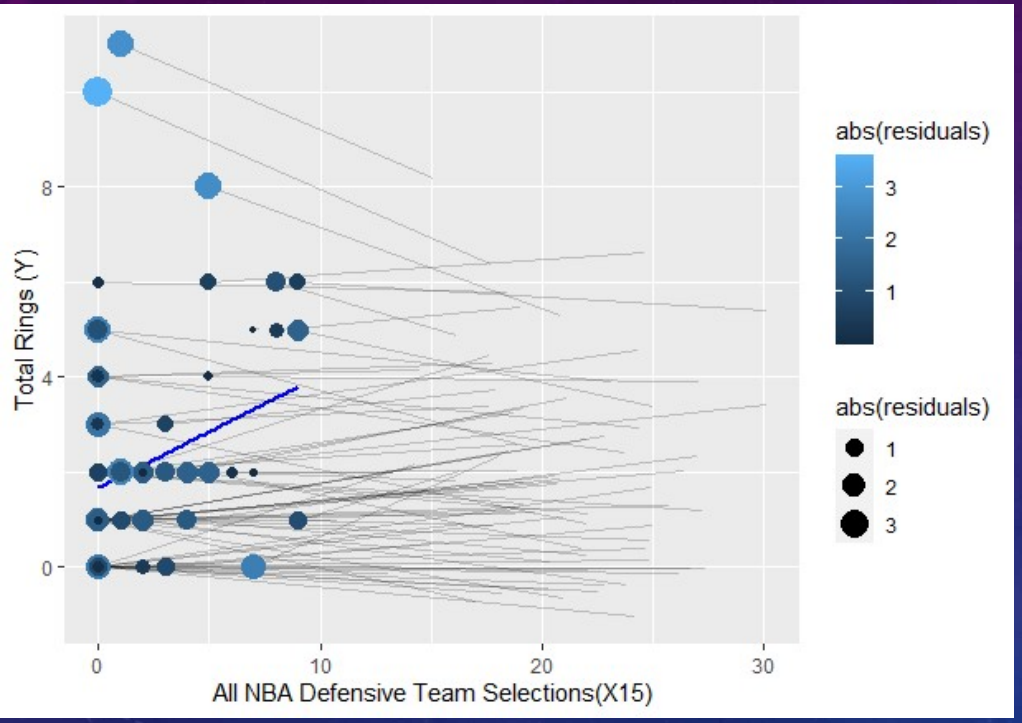


Career Wins

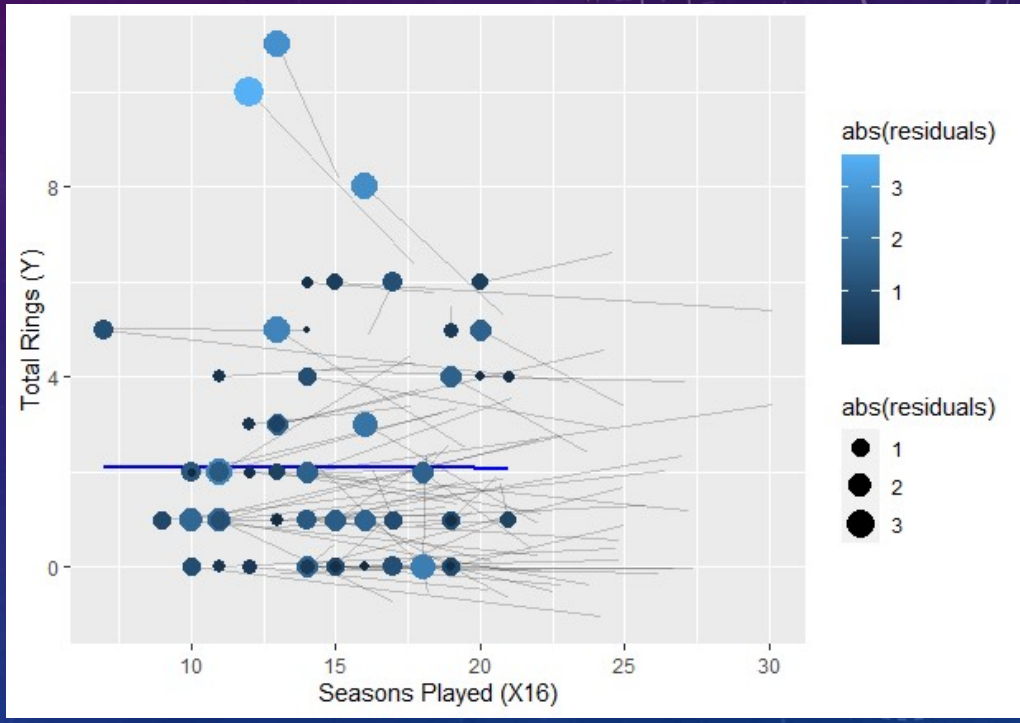


Career Losses

APPENDIX: RESIDUAL PLOT CONT.

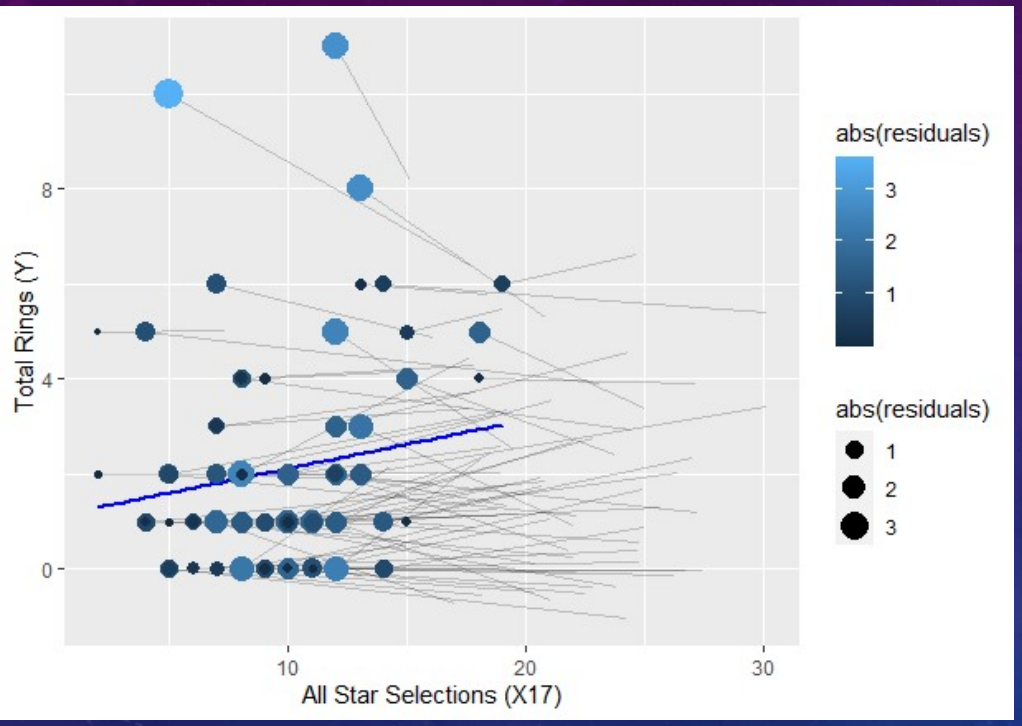


All NBA Defensive Team Selections

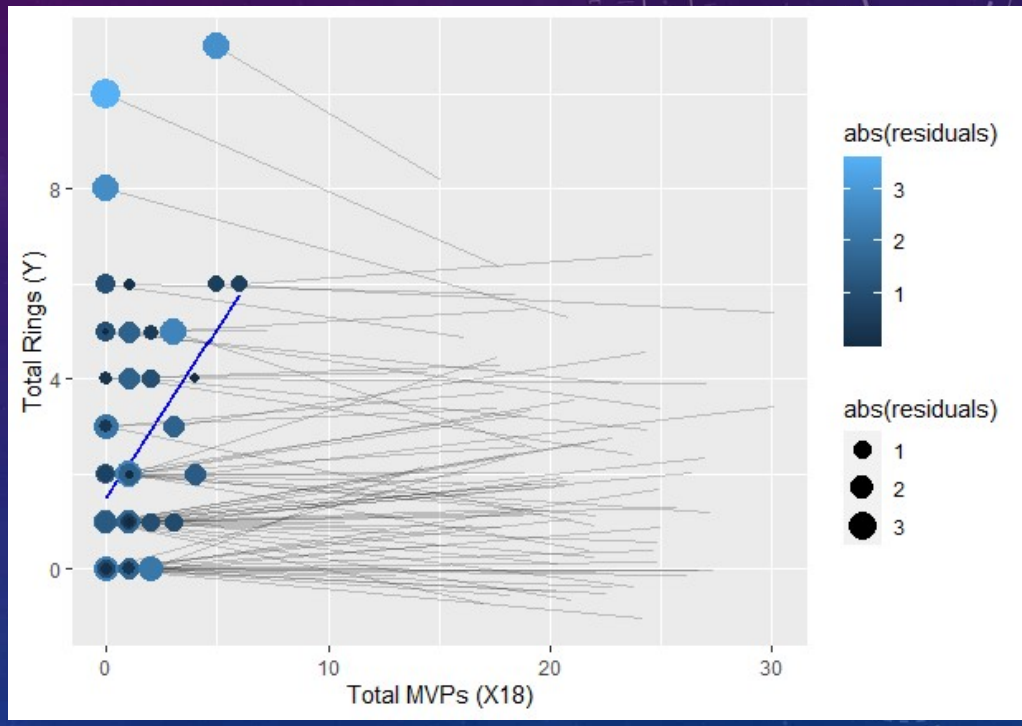


Seasons Played

APPENDIX: RESIDUAL PLOT CONT.

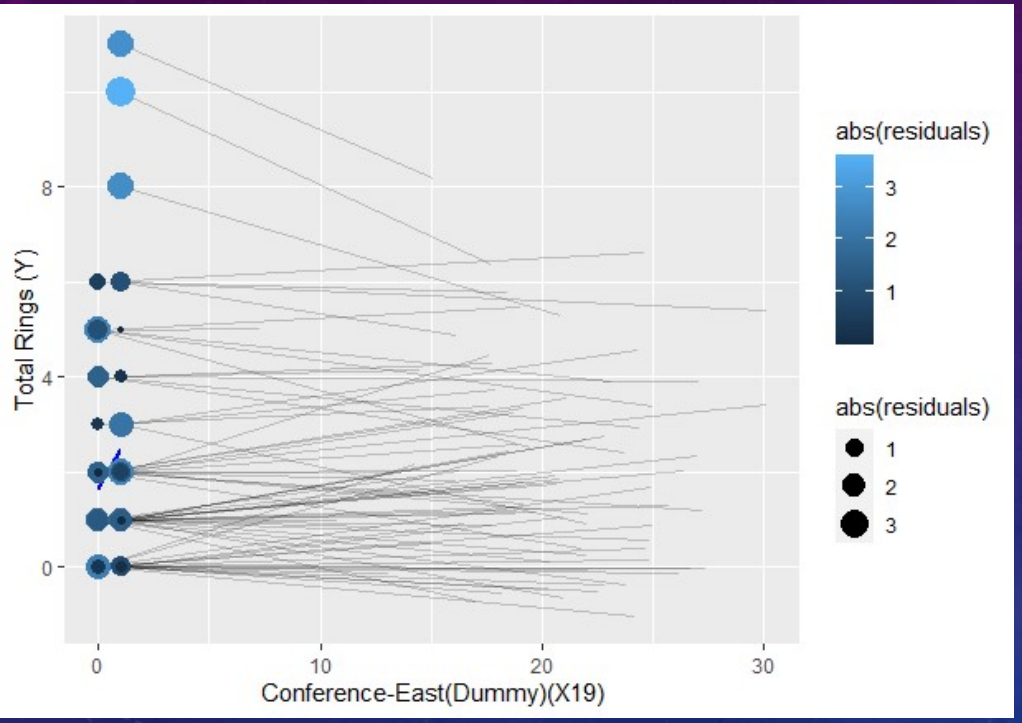


All Star Selections

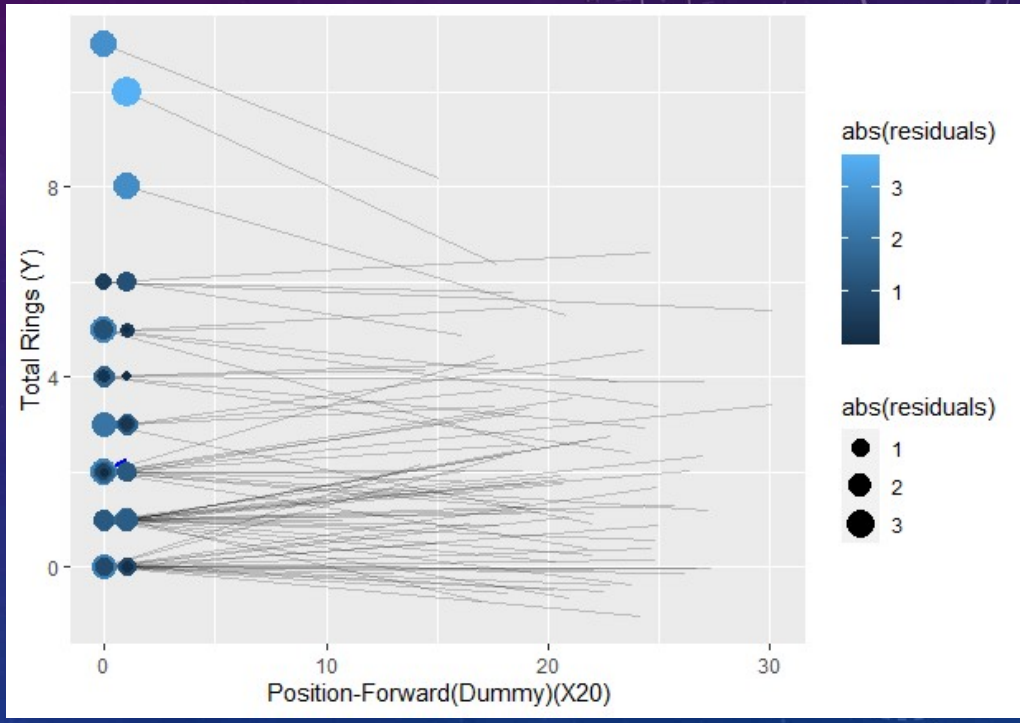


Total MVPs

APPENDIX: RESIDUAL PLOT CONT.

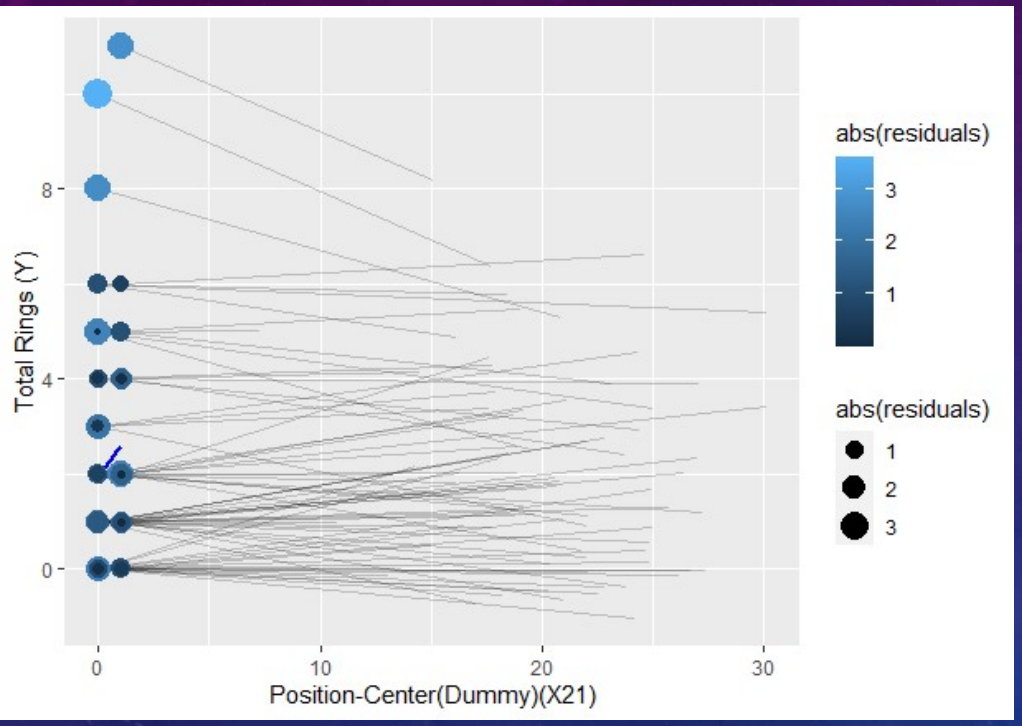


Eastern Conference Competition

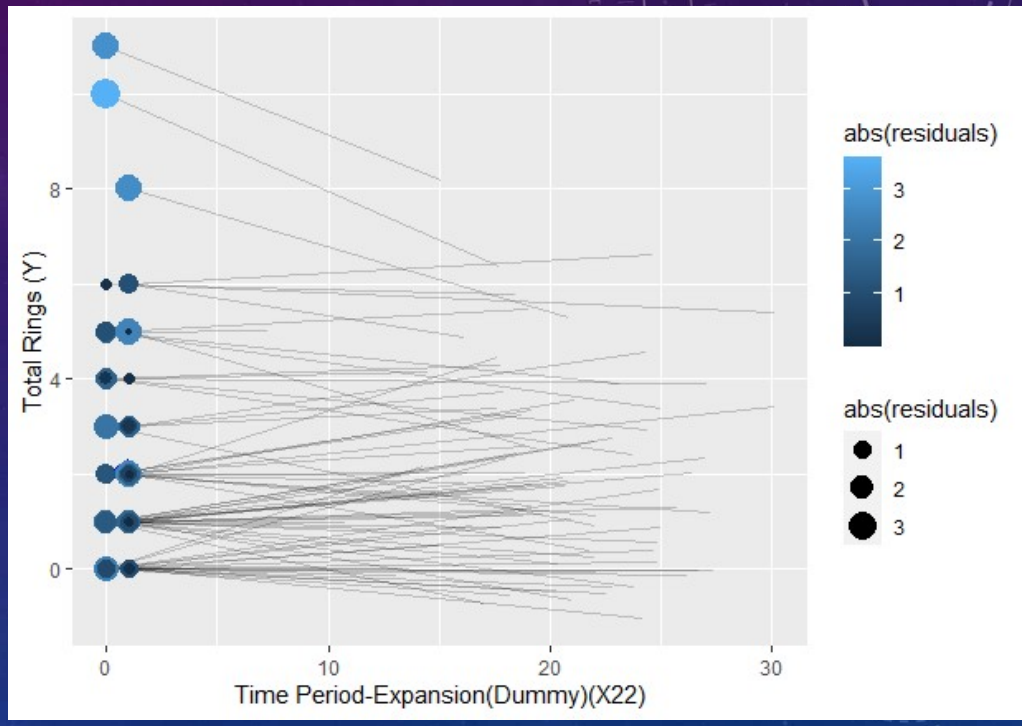


Position - Forward

APPENDIX: RESIDUAL PLOT CONT.

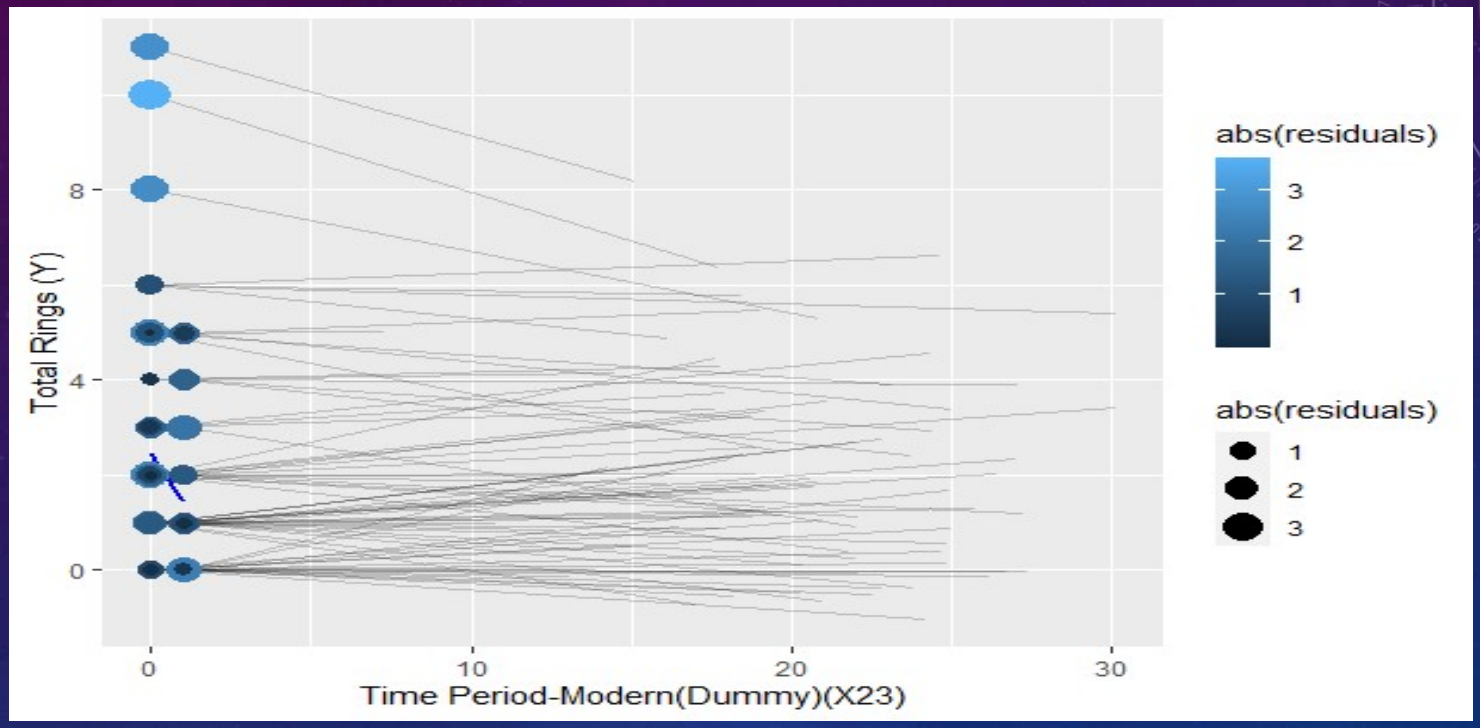


Position - Center



Time Period: Expansion

APPENDIX: RESIDUAL PLOT CONT.



Time Period: Modern

APPENDIX: R CODE

```
### Import data set with set up dummy variables ###
FPA_FinalProject_RData <- read_excel("~/FPA_FinalProject_RData.xlsx")
view(FPA_FinalProject_RData)
glimpse(FPA_FinalProject_RData)
FPA_FinalProject_RData$`Total Rings (Y)`

###OLS Regression###
model <- lm(data = FPA_FinalProject_RData, `Total Rings (Y)` ~ `Career PPG (X1)`+`Career APG (X2)`+`Career RPG (X3)`+`Career SPG (X4)`+
            `Career BPG (X5)`+`Career FG% (X6)`+`Career 3PFG% (X7)`+`Career Points High(X8)`+`Career Assists High(X9)`+`Career Rebounds High(X10)`+
            `Playoff Appearances (X11)`+ `All NBA 1st Team selections (X12)`+`Career Wins(X13)`+ `Career Losses(X14)`+
            `All NBA Defensive Team Selections(X15)`+ `Seasons Played (X16)`+`All Star Selections (X17)`+`Total MVPs (X18)`+
            `Conference-East(Dummy)(X19)`+`Position-Forward(Dummy)(X20)`+`Position-Center(Dummy)(X21)`+`Time Period-Expansion(Dummy)(X22)`+
            `Time Period-Modern(Dummy)(X23)`)
summary(model)
model_residuals <- residuals(model)
sum(model_residuals)
### sum of residuals 0 (3.080869e-15) ###
### Significant variables are FG%, Wins, Losses, Eastern Conference ###

### DW Test (two tailed) ###
durbinWatsonTest(model)
### Dstat = 1.829056 ###
### alpha = 0.05, n = 75 k = 20 du = 1.027  dL = 2.315 ###
### Dstat = 1.829 is between dU dL so the test is inconclusive ###

### BP Test ###
bptest(model)
### BP Stat = 31.411 chi square_23_0.05= 35.172 ###
### BP Stat < chi square so we do not reject H0 ###

### White test ###
white_test(model)
### White_Stat: 16.28 p value: 0.000292 ###
### Chi square_22_0.05 = 33.924 ###
### Wstat < critical value so we do not reject H0 ###
```

APPENDIX: R CODE CONT.

```
### New model with only significant data ###
model2 =lm(data = FPA_FinalProject_RData, `Total Rings (Y)`~ `Career FG% (X6)`+`Career Wins(X13)`+`Career Losses(X14)`+`Conference-East(Dummy)(X19)`)
summary(model2)
model2_residuals <- residuals(model2)
sum(model2_residuals)
### Sum of Residuals = 0 (2.331468e-15) ###
### Playing in the eastern conference became largely more significant, as did the intercept and career wins and losses.
### FG% became slightly less significant ###

### DW Test (two tailed) ###
durbinWatsonTest(model2)
### Dstat = 2.028797 dL = 1.51 dU = 1.74 N=75 K=4 ###
### Dstat> dU so we do not reject H0 ###

### BP Test ###
bptest(model2)
### BPstat = 18.685 chisquare_4_0.05 = 9.488 ###
### BPstat > chisquare so we reject H0, evidence of heteroscedasticity ###

### White Test ###
white_test(model2)
### WStat = 26.32 chisquare_3_0.05 = 7.815 ###
### Wstat > chi_square so we reject H0, more evidence of heteroscedasticity ###

### Normality/Histogram of dependent values ###
qplot(FPA_FinalProject_RData$`Total Rings (Y)`)
### Not normally distributed ###

### Residual plot vs fitted values ###
plot(model, which=1, col=c("darkblue"))
### points closely follow the curve of the line with few outliers ###
plot(model2,which=1, col=c("green"))
### point are more closely along the curve with the same few outliers ###
```

APPENDIX: R CODE CONT.

```
### Residual Analysis ###
FPA_FinalProject_RData$predicted <- predict(model)
FPA_FinalProject_RData$residuals <- residuals(model)
ggplot(FPA_FinalProject_RData, aes(x=`Career PPG (X1)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career APG (X2)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career RPG (X3)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career SPG (X4)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career BPG (X5)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career FG% (X6)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))
```

APPENDIX: R CODE CONT.

```
ggplot(FPA_FinalProject_RData, aes(x=`Career 3PFG% (X7)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career Points High(X8)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career Assists High(X9)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career Rebounds High(X10)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Playoff Appearances (X11)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`All NBA 1st Team selections (X12)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Career Wins(X13)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))
```

APPENDIX: R CODE CONT.

```
ggplot(FPA_FinalProject_RData, aes(x=`Career Losses(X14)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`All NBA Defensive Team Selections(X15)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Seasons Played (X16)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`All Star Selections (X17)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Total MVPs (X18)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Conference-East(Dummy)(X19)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Position-Forward(Dummy)(X20)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))
```

APPENDIX: R CODE CONT.

```
ggplot(FPA_FinalProject_RData, aes(x=`Position-Center(Dummy)(X21)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Time Period-Expansion(Dummy)(X22)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))

ggplot(FPA_FinalProject_RData, aes(x=`Time Period-Modern(Dummy)(X23)`, y=`Total Rings (Y)`)) +
  geom_smooth(method ="lm", se = FALSE, color="blue")+
  geom_segment(aes(xend=`Career PPG (X1)`, yend = predicted),alpha = 0.2)+
  geom_point(aes(color=abs(residuals), size=abs(residuals)))
```